

Jimmy Dagum, Jr.

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Education

University of Nevada, Las Vegas

Bachelor of Science in Computer Science

Las Vegas, NV

Expected Graduation May 2026

Technical Skills

Languages: C++, C, Python, TypeScript, JavaScript, Go, PHP, HTML, CSS, SQL, x86-64 ASM/MIPS, MATLAB

Frameworks & Libraries: Next.js, React, Zustand, TanStack Query, Prisma, Tailwind CSS, Socket.IO, Bootstrap, Framer Motion

Databases: SQL Server, MySQL, PostgreSQL

Tools / Platforms: Git, GitLab CI/CD, Bash, Proxmox, RabbitMQ (AMQP), SSH, SSMS, Quest KACE, WebSockets, SMTP, Docker

Experience

Software Engineer Intern

May 2025 – Present

JT4

Las Vegas, NV

- Architected a secure, portable-across-networks Next.js (TypeScript/Tailwind) platform—implemented authentication and authorization with network isolation and migrated SQL Server from raw SQL to Prisma (type-safe queries and schema migrations)
- Engineered feature-level caching (e.g., org chart) with in-memory hash maps, reducing redundant DB queries by 90%
- Integrated RabbitMQ and enhanced a Go microservice to fan out batched email payloads, boosting pipeline throughput by $\approx 2\times$ via per-message retries and re-queuing of failures
- Shipped a real-time, role-scoped moderator feed over WebSockets, cutting time-to-visibility for new reports from minutes to seconds—cursor-based pagination and infinite queries reduced API calls per session by $\approx 60\%$ and DB reads by $\approx 70\%$
- Delivered in weekly Agile/Scrum sprints; used GitLab merge requests with continuous peer feedback and reviews

Student Systems Administrator

Mar 2025 – May 2025

University of Nevada, Las Vegas

Las Vegas, NV (Remote)

- Decommissioned 10+ legacy VMs (Linux & Windows) via a phased, secure approach
- Developed and maintained Python scripts leveraging the KACE API, integrating them within the GitLab CI/CD pipeline (IaC) to automate regular updates for service owners

Software Engineer Intern

May 2024 – Dec 2024

NexOasis – Enterprise Security

Las Vegas, NV (Remote)

- Engineered a Proxmox-based web UI (Python, PHP, JS/Bootstrap, HTML) for automated VM certificate authority provisioning, cloning, and secure SSH file transfers—cutting manual effort by 80%
- Designed and implemented MySQL data models to manage user-specific roles and associate virtual machine IDs with parsed cron jobs, facilitating the automated cloning process and secure file transfers; improved data retrieval speed by 30%
- Implemented a Gitea Git server for 5 interns, hosting multiple repositories in a secure environment closed to outside users
- Automated the deployment and configuration of a centralized syslog server to capture all system and user update logs

Information Technology Support Consultant

Dec 2023 – May 2024

University of Nevada, Las Vegas

Las Vegas, NV

- Architected a web application using React and in-house APIs to visualize real-time solar energy data across 16 buildings and local weather conditions for UNLV
- Designed a Python-based solution to extract data from three separate databases, identifying missing key cards
- Automated data consolidation of over 10,000 work orders and related surveys using JavaScript in Google Sheets
- Provided technical support to over 250 UNLV staff in 19 buildings, including troubleshooting hardware and software issues, configuring new computers with operating systems, software applications, and networking settings
- Maintained an inventory of over 300 computers and equipment with regular audits to track the location and status of each computer

Projects

Catharsis | React (JavaScript), CSS, HTML, SQL, Git

- Created a React-based web app to host hundreds of photos and music libraries
- Established an optimized database system via PostgreSQL for efficient storage and asynchronous retrieval of user information, with the capability of thousands of users
- Integrated 2-factor authentication via SMTP to harden account access
- Reduced 10 differing states to a single global state via React Redux for streamlined music player state management

Battery Workforce Challenge (UNLV) | MATLAB, Simulink, Git

- Developed an advanced car battery cell simulation utilizing comprehensive data from the hardware team; increased model accuracy by 30% and shortened prototype development cycle by 20%
- Engineered software for assessing car battery health and charge levels using SoC and SoH algorithms via the Kalman Filter in MATLAB/Simulink, increasing the accuracy of charge level readings by 25% and reducing live code execution time by 10 seconds